

Highlights

Physics-informed LSTM and BiGRU-based approach for maneuverable satellite cluster trajectory prediction and intention recognition

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- Physics-informed conditional LSTM for satellite cluster trajectory prediction with differentiable CW-based Δv estimation
- BiGRU-based permutation-invariant network with set attention and pairwise interaction for multi-task intention recognition
- 10-channel physics-informed feature engineering covering CW residuals, orbital invariants, and inter-target geometric statistics
- Comprehensive dataset covering five maneuver intention scenarios generated via PPO-based orbital pursuit simulation

Physics-informed LSTM and BiGRU-based approach for maneuverable satellite cluster trajectory prediction and intention recognition^{*}

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ABSTRACT

This paper proposes a two-stage deep learning framework for maneuverable satellite cluster trajectory prediction and intention recognition. In the first stage, a physics-informed conditional long short-term memory (PI-LSTM) network is developed for trajectory prediction, which incorporates a differentiable Clohessy–Wiltshire (CW) inverse module to estimate velocity increments from observed states and feeds them as conditioning signals into the LSTM encoder–decoder with residual delta prediction. In the second stage, a bidirectional gated recurrent unit (BiGRU) network with permutation-invariant set attention aggregation and pairwise interaction modeling performs multi-task intention recognition, classifying the number of cluster objects (N) and regressing the global minimum distance ($d_{\min}$) and spatial dispersion angle (ϕ). A 10-channel physics-informed feature representation is constructed from CW propagation residuals, orbital invariants, and inter-target geometric statistics. The dataset comprises samples generated from reinforcement-learning-based simulations covering five maneuver intention scenarios: pursuit, reconnaissance, ambush, obstruction, and mixed strategies. Numerical results demonstrate that the PI-LSTM achieves a position RMSE of 0.148 km with strong CW self-consistency, and the BiGRU-based network attains 99.1% accuracy for N classification and an R^2 of 0.991 for $d_{\min}$ regression.

1. Introduction

The rapid advancement of space technology and the accelerating deployment of large-scale satellite constellations have fundamentally reshaped the orbital environment. The population of tracked resident space objects — including operational spacecraft, defunct satellites, and fragment debris — now numbers in the tens of thousands, substantially elevating the risk of orbital congestion and collision Carrillo, Giraldo, Martin and Valencia (2024). This increasingly crowded domain is further complicated by the growing prevalence of non-cooperative and adversarial behaviors, as multiple space actors vie for strategic advantages in orbit. Against this backdrop, the capability to forecast the trajectories and infer the operational intents of maneuverable satellite clusters has become an essential pillar of space situational awareness (SSA) and asset protection. Accurate trajectory prediction and intention recognition for such targets provide critical decision-support for collision avoidance and threat assessment, and are vital for maintaining sustainable space operations Carrillo et al. (2024).

Trajectory prediction for spacecraft has traditionally relied on physics-based propagation models. The Clohessy–Wiltshire (CW) equations Clohessy and Wiltshire (1960)

provide an analytical linearized description of relative motion in near-circular orbits and remain widely employed for orbital rendezvous guidance and proximity operations. Linear covariance techniques have been developed to analyze trajectory dispersions and navigation errors in closed-loop rendezvous scenarios Geller (2006). However, when unknown impulsive maneuvers occur, pure physics-based methods diverge rapidly from the true trajectory because the governing dynamics are no longer accurately described by the unforced CW model. To address this limitation, data-driven approaches have gained considerable attention in recent years. Deep learning models, particularly recurrent neural networks, have demonstrated remarkable capability in capturing temporal dependencies for sequential prediction tasks Hochreiter and Schmidhuber (1997); Chung, Gülçehre, Cho and Bengio (2014). Long short-term memory (LSTM) networks have been successfully applied to satellite orbit prediction and orbit determination under continuous maneuvers Zhou, Qin, Ji and Qiao (2023). Bidirectional LSTM (Bi-LSTM) architectures have demonstrated effectiveness in atmospheric drag coefficient modeling for low Earth orbit (LEO) satellites Chen, Tang, Dai, Hu, Chen, Zhang, Zhu and Li (2025) and in data-driven orbit prediction for LEO spacecraft Ai, Yue, Wang and Yue (2025).

The integration of physical knowledge with data-driven models, known as physics-informed neural networks (PINNs), has emerged as a promising paradigm for scientific machine learning Raissi, Perdikaris and Karniadakis (2019). By embedding governing equations into the learning objective, PINNs produce predictions that respect physical conservation laws while benefiting from the expressive power of neural networks. In the orbital mechanics

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domain, physics-informed approaches have been explored for orbit determination using extreme learning machines with physics-based regularization Scorsoglio, Ghilardi and Furfaro (2023). Neural ordinary differential equations (Neural ODEs) have been combined with physical models for precise LEO orbit prediction Kamiya, Etou, Alesiani and Takamoto (2023). However, existing efforts predominantly address single-satellite scenarios and lack explicit handling of multi-agent interactions and unknown maneuvers — two critical challenges when dealing with maneuverable satellite clusters where multiple spacecraft execute coordinated or independent maneuvers simultaneously.

Intention recognition for space objects constitutes an equally important and challenging problem. Early work employed control-distance metrics Holzinger, Scheeres and Alfriend (2012) and deep convolutional neural networks Linares, Furfaro and Reddy (2020) for object correlation, maneuver detection, and classification from observational data. More recently, deep neural networks have been introduced for non-cooperative space target intention recognition using CW-based maneuver geometries Sun and Dang (2023). Bidirectional gated recurrent units (BiGRU) have been employed for multi-target maneuver-mode intent recognition in cluster scenarios Tong, Cai and Hu (2024). Nevertheless, most existing studies treat intention recognition as a pure classification problem with discrete maneuver types rather than jointly predicting the continuous geometric configuration parameters — such as the number of objects in a cluster, their closest approach distance, and their spatial dispersion — that are of direct operational relevance. Furthermore, the variable number of objects in a cluster poses a permutation-invariance requirement that conventional fixed-architecture networks cannot naturally accommodate. The challenge is compounded by the diversity of possible maneuver intentions: an approaching spacecraft may be executing a pursuit maneuver for rendezvous, a reconnaissance fly-by for inspection, an ambush trajectory for covert positioning, or an obstruction path for intercept, each producing distinct yet partially overlapping kinematic signatures.

To bridge these gaps, this paper proposes a two-stage deep learning framework for maneuverable satellite cluster trajectory prediction and intention recognition. The key contributions of this work are summarized as follows:

1. **Physics-informed conditional LSTM for trajectory prediction.** A differentiable Δv estimation module, derived from the CW state transition relation, computes velocity increments from observed state sequences and feeds them as conditioning signals into the LSTM encoder-decoder. This mechanism enables the model to adapt its gating behavior according to maneuver intensity without requiring explicit maneuver labels. A physics-regularized loss function penalizes violations of CW dynamics and enforces physically plausible velocity-change magnitudes.
2. **Permutation-invariant BiGRU-based intention recognition.** A shared-weight BiGRU encoder processes per-target trajectories augmented with 10

channels of physics-informed features. Set attention aggregation (based on the Deep Sets framework Zaheer, Kottur, Ravanbakhsh, Póczos, Salakhutdinov and Smola (2017)) provides a permutation-invariant global representation, while a pairwise interaction module constructs explicit two-body relational features for geometry-sensitive predictions. Multi-task outputs jointly classify the cluster size and regress continuous geometric parameters.

3. **Multi-scenario dataset with five maneuver intention types.** A large-scale training dataset is generated via Proximal Policy Optimization (PPO) Schulman, Wolski, Dhariwal, Radford and Klimov (2017) simulations covering five distinct maneuver intention scenarios: pursuit capture, reconnaissance detection, covert ambush, trajectory obstruction, and mixed strategies. This diversity enables the learned models to generalize across different operational regimes.

The remainder of this paper is organized as follows. Section 2 formulates the problem by establishing the dynamic models, describing the multi-scenario dataset, and outlining the key assumptions. Section 3 details the physics-informed LSTM trajectory prediction algorithm. Section 4 elaborates on the BiGRU-based intention recognition architecture with physics-informed feature engineering. Section 5 presents numerical simulations, ablation studies, and discussions. Section 6 concludes the paper.

2. Problem formulation

2.1. Orbital dynamics and the Clohessy–Wiltshire model

The mission scenario addressed in this section is illustrated in Fig. 1. The scenario involves two parties: an observation spacecraft E and a non-cooperative target cluster consisting of multiple satellites T, both operating in near-circular low Earth orbit at altitude h_{orb} . Two orthogonal coordinate systems are employed to describe the orbital motion.

The Earth-centered inertial (ECI) frame, denoted as $\mathcal{F}_I : O_I - X_I Y_I Z_I$, has its origin O_I at the Earth's center. The $O_I X_I$ axis lies in the equatorial plane and points toward the mean vernal equinox at 12:00 Terrestrial Time on 1 January 2000 (J2000.0). The $O_I Z_I$ axis coincides with the Earth's rotation axis and points northward, perpendicular to the equatorial plane. The $O_I Y_I$ axis completes the right-handed orthogonal triad.

The local-vertical local-horizontal (LVLH) frame, denoted as $\mathcal{F}_L : O_L - X_L Y_L Z_L$, is centered at the own satellite (the target spacecraft), which serves as the reference point for relative motion analysis. The $O_L X_L$ axis is aligned with the radial direction, pointing from the Earth's center to the own satellite (radially outward). The $O_L Z_L$ axis is aligned with the orbital angular momentum vector, i.e., perpendicular to the orbital plane. The $O_L Y_L$ axis completes the right-handed triad, pointing approximately along the along-track (velocity) direction.

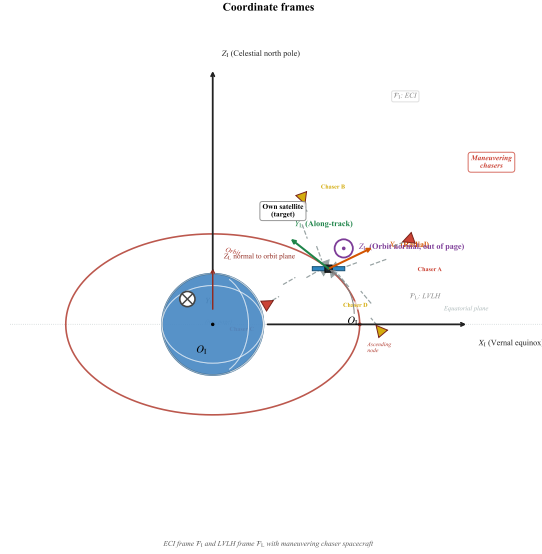


Figure 1: Coordinate frames: Earth-centered inertial (ECI) frame $\mathcal{F}_I$ and local-vertical local-horizontal (LVLH) frame $\mathcal{F}_L$, with maneuvering chaser spacecraft approaching the target satellite.

The own satellite (target) is assumed to be non-maneuvering throughout both the trajectory prediction and intention recognition processes, serving as an inertial reference for the relative motion of neighboring maneuverable spacecraft. Under these conditions, the relative motion of a chaser spacecraft with respect to the own satellite in the LVLH frame is governed by the Clohessy–Wiltshire equations Clohessy and Wiltshire (1960):

$$\ddot{x} - 2n\dot{y} - 3n^2x = 0, \quad (1)$$

$$\ddot{y} + 2n\dot{x} = 0, \quad (2)$$

$$\ddot{z} + n^2z = 0, \quad (3)$$

where x , y , z denote the radial (positive outward), along-track, and cross-track relative positions in the LVLH frame; $n = \sqrt{\mu/r^3}$ is the mean orbital angular velocity; $\mu = 398600 \text{ km}^3/\text{s}^2$ is Earth's gravitational parameter; and $r = R_E + h_{\text{orb}}$ is the orbital radius, with R_E denoting Earth's radius.

In the absence of maneuvers, the CW equations admit an analytical solution expressed through the state transition matrix $\Phi(\Delta t) \in \mathbb{R}^{6 \times 6}$. Let the state vector be $\mathbf{s} = [x, y, z, \dot{x}, \dot{y}, \dot{z}]^\top$. The free evolution over time interval Δt is then:

$$\mathbf{s}(t + \Delta t) = \Phi(\Delta t) \cdot \mathbf{s}(t). \quad (4)$$

Denoting $\tau = n \cdot \Delta t$, the non-zero entries of $\Phi(\Delta t)$ are given by:

$$\Phi_{11} = 4 - 3 \cos \tau, \quad \Phi_{14} = \frac{\sin \tau}{n},$$

$$\begin{aligned} \Phi_{15} &= \frac{2(1 - \cos \tau)}{n}, & \Phi_{22} &= 1, \\ \Phi_{21} &= 6(\sin \tau - \tau), & \Phi_{25} &= \frac{4 \sin \tau - 3\tau}{n}, \\ \Phi_{24} &= \frac{2(\cos \tau - 1)}{n}, & \Phi_{36} &= \frac{\sin \tau}{n}, \\ \Phi_{33} &= \cos \tau, & \Phi_{44} &= \cos \tau, \\ \Phi_{41} &= 3n \sin \tau, & \Phi_{54} &= -2 \sin \tau, \\ \Phi_{45} &= 2 \sin \tau, & \Phi_{55} &= 4 \cos \tau - 3, \\ \Phi_{51} &= 6n(\cos \tau - 1), & \Phi_{66} &= \cos \tau, \\ \Phi_{55} &= 4 \cos \tau - 3, & & \\ \Phi_{63} &= -n \sin \tau, & & \end{aligned} \quad (5)$$

When an impulsive velocity change (maneuver) $\Delta \mathbf{v} = [\Delta v_x, \Delta v_y, \Delta v_z]^\top$ is applied at time t , the post-maneuver state can be decomposed as:

$$\mathbf{s}(t + \Delta t) = \Phi(\Delta t) \cdot \mathbf{s}(t) + \mathbf{B}_{\text{eff}}(\Delta t) \cdot \Delta \mathbf{v}(t), \quad (6)$$

where $\mathbf{B}_{\text{eff}}(\Delta t) \in \mathbb{R}^{6 \times 3}$ consists of the velocity columns (columns 4–6) of $\Phi(\Delta t)$, representing the contribution of an instantaneous velocity impulse to the subsequent state. This linear decomposition, which is fully differentiable with respect to the state variables, serves as the cornerstone of the physics-informed approach: given consecutive observed states $\mathbf{s}_t$ and $\mathbf{s}_{t+1}$, the velocity increment can be estimated in closed form via the Moore–Penrose pseudo-inverse:

$$\Delta \mathbf{v}_t \approx \mathbf{B}_{\text{eff}}^+(\Delta t) \cdot (\mathbf{s}_{t+1} - \Phi(\Delta t) \cdot \mathbf{s}_t), \quad (7)$$

where $\mathbf{B}_{\text{eff}}^+ = (\mathbf{B}_{\text{eff}}^\top \mathbf{B}_{\text{eff}})^{-1} \mathbf{B}_{\text{eff}}^\top$ is the least-squares solution. This operation is differentiable and can therefore be embedded as a module within a neural network, enabling gradient-based learning of $\Delta \mathbf{v}$ -aware representations without requiring explicit maneuver labels.

2.2. Multi-scenario dataset generation

The dataset is constructed from orbital pursuit simulations across five distinct maneuver intention scenarios. Each scenario is modeled as a Markov decision process and solved using Proximal Policy Optimization Schulman et al. (2017). The simulation configuration is as follows. A non-maneuvering target satellite is placed at the LVLH origin. A chaser (pursuer) satellite starts from a random position on a sphere of radius d_{init} around the origin and executes impulsive orbital maneuvers to approach the target. The chaser's policy is trained using PPO with a 6-dimensional observation (relative position and velocity) and a 3-dimensional continuous action (velocity increment $\Delta \mathbf{v}$ per step, scaled such that $|\Delta \mathbf{v}| \leq \Delta v_{\text{max}}$). At each RL step, a single impulsive maneuver is followed by N_{sub} sub-steps of free CW propagation at time step Δt .

The five intention scenarios differ in their reward structures, which encode distinct operational objectives:

- **Pursuit (Capture):** The chaser receives a reward of $+R_{\text{cap}}$ upon reaching within distance d_{cap} of the target and a penalty of $-R_{\text{cap}}$ if the inter-satellite distance exceeds d_{bnd} . This incentivizes rapid approach and rendezvous.
- **Reconnaissance (Detection):** The chaser is rewarded for approaching the target to within a detection range $[d_{\text{det}}^{\min}, d_{\text{det}}^{\max}]$ without necessarily achieving rendezvous, simulating fly-by inspection or surveillance.
- **Ambush (Lurk):** The chaser maintains a holding distance within $[d_{\text{lurk}}^{\min}, d_{\text{lurk}}^{\max}]$ before executing a terminal approach, simulating covert positioning followed by a sudden engagement.
- **Obstruction (Obstruction):** The chaser maneuvers to position itself along the line-of-sight between the target and a third reference point, simulating denial of access or field-of-view obstruction.
- **Mixed (Mix):** The chaser's policy is trained with a composite reward that randomly switches between the above objectives during training, producing trajectories that blend multiple behavioral patterns.

Successful trajectories (those meeting the scenario-specific success criteria) are recorded at Δt intervals. For each scenario, approximately N_{traj} successful trajectories are collected.

2.3. Dataset construction and labels

To construct multi-target cluster scenarios, the recorded single-chaser trajectories are randomly grouped into batches of $N \in \{N_{\min}, \dots, N_{\max}\}$ targets. For each group, the final T_{total} time steps are extracted from each trajectory. If the raw trajectory length is not an exact multiple of T_{obs} , the remaining steps are padded using the CW state transition matrix $\Phi(T_{\text{pad}})$ with $T_{\text{pad}} = N_{\text{sub}} \cdot \Delta t$. The resulting T_{total} -step sequence is partitioned into an observation window $\mathbf{X}_{\text{now}} \in \mathbb{R}^{T_{\text{obs}} \times 6N}$ (the first T_{obs} steps) and a prediction target $\mathbf{X}_{\text{next}} \in \mathbb{R}^{T_{\text{pred}} \times 6N}$ (the subsequent T_{pred} steps, where $T_{\text{pred}} = T_{\text{obs}}$). For each sample, three scalar labels are computed:

- $N \in \{N_{\min}, \dots, N_{\max}\}$: the number of objects in the cluster;
- $d_{\min}$: the global minimum distance from the LVLH origin to any object over the entire T_{total} -step window;
- ϕ : the angle between the position vectors of the nearest and farthest objects from the origin at the time step of $d_{\min}$, defined as

$$\phi = \arccos \left(\frac{\mathbf{r}_{\text{nearest}} \cdot \mathbf{r}_{\text{farthest}}}{|\mathbf{r}_{\text{nearest}}| |\mathbf{r}_{\text{farthest}}|} \right) \in [0, \pi]. \quad (8)$$

The complete dataset contains N_{samples} samples. The dataset is randomly partitioned into training (ratio r_{train}), validation (ratio r_{val}), and test (ratio r_{test}) sets using a fixed random seed s .

2.4. Problem formulation

With the above setup, two interlinked tasks are defined:

Task 1 — Trajectory prediction: Given the observation window $\mathbf{X}_{\text{now}} \in \mathbb{R}^{T_{\text{obs}} \times 6N}$ of N targets (where N may range from $N_{\min}$ to $N_{\max}$), predict the future trajectory $\hat{\mathbf{X}}_{\text{next}} \in \mathbb{R}^{T_{\text{pred}} \times 6N}$. Owing to the variable N , the state matrix is zero-padded to a fixed dimension of $6N_{\max}$, with a corresponding binary mask indicating valid features.

Task 2 — Intention recognition: Given the full T_{total} -step trajectory $\mathcal{T} \in \mathbb{R}^{N \times T_{\text{total}} \times 6}$ (constructed by concatenating $\mathbf{X}_{\text{now}}$ and $\mathbf{X}_{\text{next}}$), predict the cluster configuration label $\hat{\mathbf{y}} = [N, \hat{d}_{\min}, \hat{\phi}]^T$ comprising one discrete class (N) and two continuous quantities ($d_{\min}, \phi$).

The two tasks are complementary: trajectory prediction provides the future states needed for short-term collision risk assessment, while intention recognition extracts higher-level semantic information (cluster size, geometry) that supports tactical decision-making. Both tasks must handle the permutation invariance of targets (the labeling of individual objects in the cluster is arbitrary) and the variable number of objects per sample.

3. Physics-informed LSTM trajectory prediction

3.1. Model architecture overview

The trajectory prediction task requires mapping a short observation window (T_{obs} steps, Δt per step) to an equally long prediction horizon (T_{pred} steps, with $T_{\text{pred}} = T_{\text{obs}}$). Given the short-sequence regime and the Markovian nature of orbital dynamics, an LSTM encoder-decoder with residual delta prediction provides a natural inductive bias. Building on this backbone, two physically motivated enhancements are introduced: (1) a differentiable Δv estimation module that provides maneuver-aware conditioning signals to the LSTM, and (2) a physics-regularized loss function that enforces consistency with CW dynamics.

3.2. Differentiable Δv estimation

Given an input state sequence, the Δv estimation module computes the velocity increment between each pair of consecutive states using the CW inverse relation in Eq. (7). For each target agent a with its state sequence $\mathbf{s}_{1:T}^{(a)}$, the module computes:

$$\Delta \mathbf{v}_t^{(a)} = \mathbf{B}_{\text{eff}}^+(\Delta t) \cdot (\mathbf{s}_{t+1}^{(a)} - \Phi(\Delta t) \cdot \mathbf{s}_t^{(a)}), \quad t = 1, \dots, T-1, \quad (9)$$

where Δt is the simulation time step, and $\Phi(\Delta t)$ and $\mathbf{B}_{\text{eff}}(\Delta t)$ are pre-computed constants registered as non-trainable buffers. Since both the matrix multiplication and the pseudo-inverse are differentiable operations, gradient flows through this module during backpropagation.

The per-agent Δv estimates are aggregated into a global maneuver signature by averaging over all valid agents (handling variable N via masking):

$$\Delta \bar{v}_t = \frac{1}{\sum_a \mathbb{I}[a \text{ valid}]} \sum_{a: \text{valid}} \Delta v_t^{(a)} \in \mathbb{R}^3. \quad (10)$$

3.3. Condition vector construction

The condition vector $c_t \in \mathbb{R}^{d_c}$ is constructed by summing two components:

$$c_t = \text{Embed}(m_t) + \text{MLP}_{dv}(\Delta \bar{v}_t), \quad (11)$$

where $\text{Embed}(m_t)$ is a learnable embedding of the maneuver mode label $m_t \in \{0, 1\}$ (non-maneuvering vs. maneuvering) with $2 \times d_c$ entries initialized with orthogonal weights. MLP_{dv} is a two-layer perceptron with GELU non-linearity that maps the 3-dimensional global $\Delta \bar{v}_t$ to the d_c -dimensional embedding space. Summing these two components encourages the model to learn a joint manifold where mode identity and maneuver magnitude jointly condition the downstream LSTM dynamics.

3.4. Encoder-decoder LSTM with residual prediction

The encoder LSTM receives an expanded input that concatenates each agent's 6-dimensional state with the shared d_c -dimensional condition vector, producing a $N_{\max} \times (6 + d_c)$ -dimensional input vector per time step (with invalid agent slots zeroed via the mask). The encoder consists of a n_{lstm} -layer LSTM with hidden size H_{lstm} and dropout rate p_{drop} applied between layers. The LSTM weights follow a specialized initialization: input-to-hidden weights use Xavier uniform, hidden-to-hidden weights use orthogonal initialization, and the forget gate bias is set to 1 to mitigate vanishing gradients during early training.

The decoder adopts a residual delta prediction strategy. Rather than directly generating future states, the model predicts deviations δ_t from a persistence forecast (repeating the last observed state $s_{T_{\text{obs}}}$ across all T_{pred} steps):

$$\hat{s}_{T_{\text{obs}}+k} = s_{T_{\text{obs}}} + \hat{\delta}_k, \quad k = 1, \dots, T_{\text{pred}}. \quad (12)$$

This delta formulation guarantees that the model output cannot be worse than the naive baseline and significantly reduces the learning difficulty Zhou et al. (2023). The decoder state is initialized from the encoder's final hidden state $(h_{T_{\text{obs}}}^{\text{enc}}, c_{T_{\text{obs}}}^{\text{enc}})$. A context vector $z = \text{MLP}_{\text{ctx}}(h_{T_{\text{obs}}}^{\text{enc}})$ is computed and concatenated with each step's delta prediction (augmented with the decoder-step condition vector) to form the decoder LSTM input. The decoder employs an identical n_{lstm} -layer LSTM configuration (hidden size H_{lstm}). At each time step k , the decoder hidden output passes through a two-layer output projection (GELU + Dropout + Linear) to produce $\hat{\delta}_k \in \mathbb{R}^{6N_{\max}}$. A separate Δv predictor head — a two-layer MLP ($H_{\text{lstm}}/2$ hidden units, GELU) — estimates $\Delta \hat{v}_k$ from the decoder hidden state at each output step for physics loss computation.

During training, teacher forcing is applied with a linearly decaying probability: starting at 1.0 (complete reliance on ground-truth deltas) and decaying to 0 over the first E_{tf} epochs, ensuring a smooth transition from guided learning to autonomous generation. At inference time, teacher forcing is disabled. The complete model contains approximately $N_{\text{param}}^{\text{traj}}$ trainable parameters. The overall architecture is illustrated in Fig. 2.

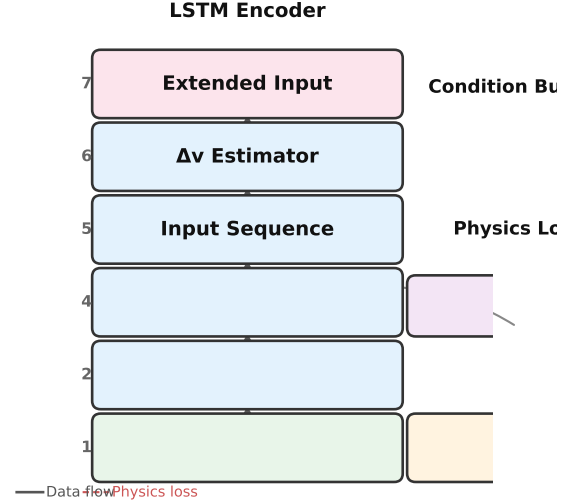


Figure 2: Architecture of the physics-informed conditional LSTM (PI-LSTM) for trajectory prediction. The framework comprises eight key components: (1) input trajectory sequence, (2) differentiable Δv estimator via CW inverse, (3) condition builder with mode embedding, (4) extended input combining state and condition features, (5) LSTM encoder, (6) LSTM decoder with residual delta prediction, (7) predicted trajectory output, and (8) physics-based regularization enforcing CW consistency.

3.5. Physics-based loss regularization

The training objective comprises three terms:

$$\mathcal{L}_{\text{pred}} = \lambda_{\text{pred}} \mathcal{L}_{\text{MSE}} + \lambda_{\text{phy}} \mathcal{L}_{\text{physics}} + \lambda_{\text{mode}} \mathcal{L}_{\text{mode}}. \quad (13)$$

3.5.1. Prediction loss

$\mathcal{L}_{\text{MSE}}$ is a masked mean squared error over valid features:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{\sum \mathbf{M}} \sum_{i,j: \mathbf{M}_{ij}=1} (\hat{X}_{\text{next}}^{(i,j)} - X_{\text{next}}^{(i,j)})^2, \quad (14)$$

where $\mathbf{M} \in \{0, 1\}^{B \times 6N_{\max}}$ is the batch mask indicating valid (non-padding) feature dimensions.

3.5.2. Physics consistency loss

$\mathcal{L}_{\text{physics}}$ enforces CW self-consistency on the predicted sequences:

$$\mathcal{L}_{\text{physics}} = \underbrace{\frac{1}{T-1} \sum_{t=1}^{T-1} \|s_{t+1} - \Phi(\Delta t) \cdot s_t\|^2}_{\text{CW free-evolution residual}} + \underbrace{\alpha \cdot \frac{1}{T} \sum_t \max(0, |\Delta v_t| - \Delta v_{\max})^2}_{\Delta v \text{ boundary penalty}}, \quad (15)$$

where $\Delta v_{\max}$ is the physically plausible maneuver bound. The CW residual term penalizes trajectories whose adjacent steps deviate from the expected free-evolution pattern; the boundary penalty discourages the model from estimating unrealistically large velocity changes.

3.5.3. Mode consistency loss

$\mathcal{L}_{\text{mode}}$ aligns the model's own Δv estimates (from the decoder head) with those computed by the Δv estimator module from the input sequence:

$$\mathcal{L}_{\text{mode}} = \|\Delta \hat{v}^{\text{dec}} - \Delta v^{\text{CW}}\|^2, \quad (16)$$

encouraging consistency between the decoder's internal maneuver representation and the physically grounded CW estimates.

3.5.4. Loss weight scheduling

The loss weights follow a warmup schedule to avoid early interference with the primary prediction task. For the first E_{warmup} epochs, only $\mathcal{L}_{\text{MSE}}$ is active. From epoch E_{warmup} to $E_{\text{warmup}} + E_{\text{ramp}}$, λ_{phy} linearly increases from 0 to $\lambda_{\text{phy}}^{\max}$ and λ_{mode} from 0 to $\lambda_{\text{mode}}^{\max}$, after which they remain constant. This staged training strategy allows the model to first learn basic trajectory dynamics before being subjected to physics-based constraints.

3.6. Training configuration

The model is trained using the AdamW optimizer Loshchilov and Hutter (2019) with a peak learning rate η_{traj} , weight decay $\lambda_{\text{wd}}^{\text{traj}}$, and batch size B_{traj} . The learning rate follows a cosine warmup schedule ($E_{\text{warmup}}^{\text{lr}}$ epochs to reach the peak) followed by cosine annealing to a minimum of $\eta_{\text{min}}^{\text{traj}}$. Gradient clipping with ℓ_2 -norm threshold g_{clip} is applied to stabilize training. Early stopping with patience of P_{traj} epochs on the validation prediction loss is employed, with a maximum of E_{traj} epochs. All features are Z-score normalized using statistics computed solely on the training set.

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1: Input: Observed trajectory  $S_{\text{obs}} = \{s_1, \dots, s_{T_{\text{obs}}}\}$ ,  $s_t \in \mathbb{R}^6$ 
2: Output: Predicted trajectory  $S_{\text{pred}} = \{\hat{s}_{T_{\text{obs}}+1}, \dots, \hat{s}_{T_{\text{obs}}+T_{\text{pred}}}\}$ 
3: # Stage 1: Differentiable  $\Delta v$  estimation
4: for  $t = 1$  to  $T_{\text{obs}} - 1$  do
5:    $\Delta v_t \leftarrow \text{CWInverse}(s_t, s_{t+1})$ 
6:    $c_t \leftarrow \text{MLP}_{\text{cond}}([\text{ModeEmbed}(S_{\text{obs}}), \Delta v_t])$ 
7: end for
8:  $c_0 \leftarrow \mathbf{0}$  # Initial zero-padded condition
9: # Stage 2: LSTM encoder
10: for  $t = 1$  to  $T_{\text{obs}}$  do
11:    $x_t \leftarrow [s_t; c_{t-1}]$ 
12:    $(h_t^{\text{enc}}, c_t^{\text{enc}}) \leftarrow \text{LSTM}_{\text{enc}}(x_t, h_{t-1}^{\text{enc}}, c_{t-1}^{\text{enc}})$ 
13: end for
14:  $z \leftarrow \text{MLP}_{\text{ctx}}(h_{T_{\text{obs}}}^{\text{enc}})$  # Context vector for decoder
15:  $(h_0^{\text{dec}}, c_0^{\text{dec}}) \leftarrow (h_{T_{\text{obs}}}^{\text{enc}}, c_{T_{\text{obs}}}^{\text{enc}})$ 
16: # Stage 3: LSTM decoder with residual prediction
17: for  $k = 1$  to  $T_{\text{pred}}$  do
18:   if training and teacher forcing then
19:      $\tilde{\delta}_k \leftarrow \delta_k^{\text{gt}}$ 
20:   else
21:      $\tilde{\delta}_k \leftarrow \hat{\delta}_{k-1}$ 
22:   end if
23:    $u_k \leftarrow [z; \tilde{\delta}_k; c_{T_{\text{obs}}+k-1}]$ 
24:    $(h_k^{\text{dec}}, c_k^{\text{dec}}) \leftarrow \text{LSTM}_{\text{dec}}(u_k, h_{k-1}^{\text{dec}}, c_{k-1}^{\text{dec}})$ 
25:    $\hat{\delta}_k \leftarrow \text{MLP}_{\text{out}}(h_k^{\text{dec}})$  # Residual delta prediction
26:    $\hat{s}_{T_{\text{obs}}+k} \leftarrow s_{T_{\text{obs}}} + \hat{\delta}_k$ 
27:    $\Delta \hat{v}_k \leftarrow \text{MLP}_{\text{dv}}(h_k^{\text{dec}})$  #  $\Delta v$  for physics loss
28: end for
29: # Stage 4: Physics-based regularization
30:  $\mathcal{L}_{\text{pred}} \leftarrow \lambda_{\text{pred}} \mathcal{L}_{\text{MSE}} + \lambda_{\text{phy}} \mathcal{L}_{\text{physics}} + \lambda_{\text{mode}} \mathcal{L}_{\text{mode}}$ 
31: return  $\hat{S}_{\text{pred}}$ 

```

Figure 3: PI-LSTM trajectory prediction with physics-informed conditional LSTM. The algorithm comprises four stages: differentiable Δv estimation via CW inverse (5), condition vector construction (6), encoder-decoder LSTM with extended input (11)–(12), autoregressive residual prediction with teacher forcing (19)–(27), and physics-based multi-objective optimization (30).

4. BiGRU-based intention recognition

4.1. Design principle: physics-guided representation learning

The intention recognition task maps a T_{total} -step trajectory of N targets to a configuration label $y = [N, d_{\min}, \phi]^T$. Unlike trajectory prediction — where the output is a dense spatial-temporal sequence — intention recognition demands a compact, permutation-invariant representation that aggregates information across a variable number of targets and time steps into three semantically distinct scalar outputs. The proposed approach constructs physically meaningful features from the trajectory using the CW model and orbital mechanics, then learns a permutation-invariant architecture that respects the set structure of the input.

4.2. Physics-informed feature engineering

For each target, the 6-dimensional state vector $[x, y, z, \dot{x}, \dot{y}, \dot{z}]$ is augmented with d_{phys} physics-based feature channels, computed at every time step. The feature computation relies on the CW state transition matrix $\Phi(\Delta t)$ and fundamental orbital relations, both implemented without learnable parameters.

4.2.1. Per-target features (d_{phys} channels)

The d_{phys} -channel per-target feature vector is defined as follows:

1. **Distance from origin:** $r = \sqrt{x^2 + y^2 + z^2}$. This directly informs the d_{min} prediction.
2. **Speed:** $v = \sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2}$.
3. **Radial velocity:** $v_r = (\mathbf{r} \cdot \mathbf{v})/r$, characterizing whether the target moves toward or away from the origin.
4. **Angular momentum magnitude:** $h = |\mathbf{r} \times \mathbf{v}|$, an invariant under CW dynamics in the absence of along-track perturbations.
5. **Orbital energy:** $E = v^2/2 - \mu/r$, a conserved quantity under Keplerian motion.
6. **CW propagation residual (position norm):** For each of the T_{pred} future steps (steps $T_{\text{obs}} + 1$ to T_{total}), the final observed state $\mathbf{s}_{T_{\text{obs}}}$ is propagated forward using $\Phi(\Delta t)$ and the position norm of the residual is computed as $|\mathbf{r}_i^{\text{actual}} - \mathbf{r}_i^{\text{CW}}|$. For the first T_{obs} steps (the observation window), these features are set to zero.
7. **CW propagation residual (velocity norm):** Same computation for the velocity norm.
8. **-10. Inter-target cosine angle statistics:** For each target i , the cosine of the angle between its position vector $\mathbf{r}_i$ and every other target's position vector $\mathbf{r}_j$ is computed: $\cos \theta_{ij} = (\mathbf{r}_i \cdot \mathbf{r}_j) / (|\mathbf{r}_i| |\mathbf{r}_j|)$. The minimum, maximum, and mean of these cosine values (over $j \neq i$) provide three channels that directly inform the ϕ prediction task.

4.2.2. Global features

In addition to the per-target features, d_{global} global scalar features are computed for each sample, capturing aggregate statistics that are independent of the target ordering:

- Minimum and maximum of all per-target distances over all time steps;
- Mean, minimum, and mean-pairwise-minimum of inter-target distances;
- Mean inter-target cosine angle;
- Mean and standard deviation of all per-target speeds;
- Standard deviation of per-target distances at the time step of global minimum distance.

All features are Z-score normalized using statistics computed exclusively on the training set. Position and velocity components are normalized separately due to their differing physical units.

4.3. Permutation-invariant network architecture

The variable number of targets ($N \in \{N_{\text{min}}, \dots, N_{\text{max}}\}$) and the arbitrariness of their ordering demand a permutation-invariant architecture. A Deep Sets-style Zaheer et al. (2017) design is adopted, with shared per-element encoding followed by set-level aggregation, augmented with explicit pairwise interaction modeling.

4.3.1. Per-target encoder (shared BiGRU)

Each target's trajectory, represented as a $(T_{\text{total}}, d_{\text{in}}^{\text{rec}})$ matrix (6 raw states + d_{phys} physical features, where $d_{\text{in}}^{\text{rec}} = 6 + d_{\text{phys}}$), is processed by an identical encoder with shared weights. A two-layer 1D convolutional projection (kernel size k_{conv} , padding 1, ReLU activation) maps the $d_{\text{in}}^{\text{rec}}$ input channels to a d_{conv} -dimensional hidden representation, followed by a n_{gru} -layer bidirectional GRU (BiGRU) Chung et al. (2014) with hidden size H_{gru} . The final forward and backward hidden states are concatenated to produce a fixed d_{emb} -dimensional embedding per target (where $d_{\text{emb}} = 2H_{\text{gru}}$). The use of BiGRU rather than Transformer follows from the observation that for short sequences, the recurrent inductive bias is more parameter-efficient and empirically more effective for this task.

4.3.2. Set attention aggregation

To aggregate the variable-size set of target embeddings into a fixed-dimensional global representation, multi-head attention pooling is employed Lee, Lee, Kim, Kosiorek, Choi and Teh (2019). A learnable seed vector $\mathbf{q} \in \mathbb{R}^{d_{\text{emb}}}$ (initialized with small variance) serves as the query, attending to the N target embeddings as keys and values through a n_{head} -head attention mechanism:

$$\mathbf{g} = \text{LayerNorm}(\mathbf{q} + \text{MultiHeadAttn}(\mathbf{q}, \{\mathbf{e}_i\}_{i=1}^N, \{\mathbf{e}_i\}_{i=1}^N)). \quad (17)$$

The attention operation is inherently permutation-invariant with respect to the ordering of the target embeddings. A valid-target mask ensures that padded (non-existent) targets are excluded from the attention computation.

4.3.3. Pairwise interaction module

The set aggregation produces a global representation suitable for predicting N and d_{min} , but the angle ϕ depends on the relative geometry between two specific targets (the nearest and farthest from the origin at the d_{min} time step). To provide explicit pairwise relational features, a combined representation is constructed for each pair of valid targets (i, j) :

$$\mathbf{p}_{ij} = [\mathbf{e}_i; \mathbf{e}_j; \mathbf{e}_i \odot \mathbf{e}_j; \mathbf{e}_i - \mathbf{e}_j] \in \mathbb{R}^{4d_{\text{emb}}}, \quad (18)$$

where $[\cdot; \cdot]$ denotes concatenation and $\odot$ is element-wise multiplication. A shared two-layer MLP ($d_{\text{pair}}^{\text{hid}}$ hidden units, ReLU) processes each pair, and max-pooling across all pairs

yields a single $d_{\text{pair}}^{\text{out}}$ -dimensional pairwise feature vector. This max-pooling operation is permutation-invariant and robust to the variable number of targets.

4.3.4. Multi-task prediction heads

The final prediction layer consists of three specialized heads sharing a common d_{emb} -to- d_{shared} -dimensional projection (ReLU + Dropout):

- **N classification head:** A linear layer mapping d_{shared} dimensions to K_N logits (classes $N \in \{N_{\min}, \dots, N_{\max}\}$, where $K_N = N_{\max} - N_{\min} + 1$), trained with cross-entropy loss.
- **$d_{\min}$ regression head:** A two-layer MLP ($d_{\text{shared}} \rightarrow d_{\text{shared}}/2 \rightarrow 1$) trained with Huber loss (threshold δ_{huber}) for robustness to outliers.
- **ϕ regression head:** A three-layer MLP receiving the concatenation of the shared d_{shared} -dimensional feature and the $d_{\text{pair}}^{\text{out}}$ -dimensional pairwise interaction feature. Trained with a combined loss: $\mathcal{L}_{\phi} = \text{MSE}(\hat{\phi}, \phi) + \alpha_{\cos} \cdot (1 - \cos(\pi(\hat{\phi} - \phi)))$, where the cosine term provides periodic awareness of the circular nature of angular quantities.

The total multi-task loss is:

$$\mathcal{L}_{\text{intent}} = \lambda_N \mathcal{L}_{\text{CE}} + \lambda_d \mathcal{L}_{\text{Huber}} + \lambda_{\phi} \mathcal{L}_{\phi}, \quad (19)$$

with task-specific weights $\lambda_N, \lambda_d, \lambda_{\phi}$. The higher weight on ϕ reflects its greater learning difficulty. The complete model contains approximately $N_{\text{param}}^{\text{rec}}$ trainable parameters.

4.4. Training configuration

The model is trained with AdamW (learning rate η_{rec} , weight decay $\lambda_{\text{wd}}^{\text{rec}}$) and a CosineAnnealingWarmRestarts scheduler (initial period T_0 , multiplier T_{mult}). Batch size is B_{rec} . Dropout of $p_{\text{drop}}^{\text{rec}}$ is applied throughout. Gradient clipping with ℓ_2 -norm threshold g_{clip} is used. Early stopping with patience of P_{rec} epochs on validation loss terminates training, with a maximum of E_{rec} epochs.

```

1: Input: Multi-target trajectories  $\{S_1, S_2, \dots, S_N\}$ ,  $S_i \in \mathbb{R}^{T_{\text{obs}} \times 6}$ ,  $N \in \{2, 3, 4\}$ 
2: Output: Predicted cluster size  $\hat{N}$ , global minimum distance  $\hat{d}_{\min}$ , dispersion angle  $\hat{\phi}$ 
3: # Stage 1: Physics-informed feature extraction
4: for each target  $i = 1$  to  $N$  do
5:    $F_i \leftarrow \text{PhysFeatures}(S_i)$ 
   # 10-channel features:  $r, v, v_r, h, E$ , CW residuals, cos statistics
6:    $X_i \leftarrow [S_i; F_i]$ 
   # Augment state: ( $T_{\text{obs}} \times 16$ )
7: end for
8: # Stage 2: Per-target BiGRU encoding
9: for each target  $i = 1$  to  $N$  do
10:   $e_i \leftarrow \text{BiGRU}_{\text{enc}}(X_i) \in \mathbb{R}^{256}$ 
11: end for
12: # Stage 3: Set attention aggregation (Deep Sets)
13:  $K \leftarrow \text{MLP}_k(\{e_1, \dots, e_N\})$ ,
    $V \leftarrow \text{MLP}_v(\{e_1, \dots, e_N\})$ 
14:  $q \leftarrow \text{LearnableSeedQuery}$  # Learnable seed query vector
15:  $\alpha \leftarrow \text{softmax}(q^T K / \sqrt{d})$  # Attention over  $N$  targets
16:  $h_{\text{global}} \leftarrow \sum_{i=1}^N \alpha_i v_i$ 
17: # Stage 4: Pairwise interaction modeling
18: for each pair  $(i, j)$  with  $i < j$  do
19:   $p_{ij} \leftarrow \text{MLP}_{\text{pair}}([e_i; e_j; |e_i - e_j|; e_i \odot e_j])$ 
20: end for
21:  $h_{\text{pair}} \leftarrow \text{MaxPool}(\{p_{ij}\}) \in \mathbb{R}^{256}$ 
22:  $h \leftarrow [h_{\text{global}}; h_{\text{pair}}] \in \mathbb{R}^{512}$ 
23: # Stage 5: Multi-task prediction heads
24:  $\hat{N} \leftarrow \arg \max (\text{MLP}_{\text{cls}}(h))$  # 3-class:  $N = 2, 3, 4$ 
25:  $\hat{d}_{\min} \leftarrow \text{MLP}_{d_{\min}}(h)$  # Huber regression
26:  $\hat{\phi} \leftarrow \text{MLP}_{\phi}(h)$  # MSE + cosine penalty
27: return  $\hat{N}, \hat{d}_{\min}, \hat{\phi}$ 
    
```

Figure 4: BiGRU-based intention recognition with permutation-invariant set attention and pairwise interaction. The pipeline covers: physics-informed feature extraction (5)–(6), per-target BiGRU encoding (10), set attention aggregation (13)–(16), pairwise interaction modeling (19)–(21), and multi-task prediction (24)–(26).

5. Numerical simulations and analysis

5.1. Experimental setup and model parameters

All experiments are conducted on a single NVIDIA GPU with PyTorch 2.x. The concrete numerical values of all symbolic parameters introduced in the theoretical development (Sections 2–4) are summarized below. Unless otherwise noted, these values are used throughout all experiments.

5.1.1. Simulation and dataset parameters

Table 1 lists the orbital environment, RL simulation, and dataset construction parameters, together with the scenario-specific reward distances.

5.1.2. Model and training hyperparameters

Tables 2 and 3 list the architecture and training hyperparameters for the PI-LSTM trajectory prediction model and the BiGRU intention recognition model, respectively. These

Table 1

Simulation environment and dataset parameters.

Parameter (Symbol)	Value
<i>Orbital environment</i>	
Orbital altitude h_{orb}	480 km
Orbital radius $r = R_E + h_{\text{orb}}$	6851 km
Earth gravitational parameter μ	$3.986 \times 10^5 \text{ km}^3/\text{s}^2$
<i>RL simulation</i>	
Time step Δt	1 s
CW sub-steps per RL step N_{sub}	60
Max velocity increment Δv_{max}	$3 \times 10^{-3} \text{ km/s}$ (3 m/s)
Initial chaser distance d_{init}	[100, 120] km
<i>Scenario-specific distances</i>	
Capture success distance d_{cap}	5 km, reward $R_{\text{cap}} = 100$
Boundary penalty distance d_{bnd}	200 km
Detection range $[d_{\text{det}}^{\min}, d_{\text{det}}^{\max}]$	[10, 30] km
Lurk holding range $[d_{\text{lurk}}^{\min}, d_{\text{lurk}}^{\max}]$	[30, 80] km
<i>Dataset construction</i>	
Trajectories per scenario N_{traj}	$\approx 4,000\text{--}5,000$
Cluster size $N \in \{N_{\min}, \dots, N_{\max}\}$	{2, 3, 4}
Total sequence length T_{total} (steps)	20
Observation window T_{obs} (steps)	10
Prediction horizon T_{pred} (steps)	10
Padding time $T_{\text{pad}} = N_{\text{sub}} \cdot \Delta t$	60 s
Total samples N_{samples}	23,787
Train / Val / Test split	70% / 15% / 15%
$(r_{\text{train}}, r_{\text{val}}, r_{\text{test}})$	
Random seed s	42
Sample distribution: $N = 2 /$	46% / 31% / 23%
$N = 3 / N = 4$	

values instantiate the symbolic quantities used in Sections 3 and 4.

All features are Z-score normalized using statistics computed exclusively on the training set. Position and velocity components are normalized separately due to their differing physical units.

5.2. Trajectory prediction results

5.2.1. Prediction accuracy

The trained PI-LSTM model is evaluated on the held-out test set. All metrics are computed in the original physical units (positions in km, velocities in km/s) after inverse-transforming the normalized predictions. Table 4 summarizes the prediction performance.

The model achieves sub-100-meter mean absolute position error and sub-2 m/s mean absolute velocity error over the 10-step prediction horizon, demonstrating that the physics-informed conditional architecture successfully captures the underlying orbital dynamics from only 10 seconds of observation.

5.2.2. Physical consistency analysis

Two physics-based metrics are computed on the test set to assess whether the model's predictions respect the governing CW dynamics:

Table 2

PI-LSTM trajectory prediction model: architecture and training hyperparameters.

Parameter (Symbol)	Value
<i>Architecture</i>	
Condition vector dimension d_c	8
LSTM layers n_{lstm}	3
LSTM hidden size H_{lstm}	256
Input dimension per step $N_{\text{max}} \times (6 + d_c)$	$4 \times 14 = 56$
Output dimension $\hat{\delta}_k$	$6N_{\text{max}} = 24$
Dropout rate p_{drop}	0.15
Trainable parameters $N_{\text{param}}^{\text{traj}}$	$\approx 2.9 \times 10^6$
<i>Training</i>	
Optimizer	AdamW
Peak learning rate η_{traj}	10^{-3}
Minimum learning rate $\eta_{\text{min}}^{\text{traj}}$	10^{-6}
Weight decay $\lambda_{\text{wd}}^{\text{traj}}$	10^{-5}
Batch size B_{traj}	128
LR warmup epochs $E_{\text{warmup}}^{\text{lr}}$	5
Gradient clipping ℓ_2 threshold g_{clip}	1.0
Max epochs E_{traj}	300
Early stopping patience P_{traj}	50
<i>Physics loss schedule</i>	
Prediction-only warmup epochs E_{warmup}	10
Linear ramp duration E_{ramp}	20 (epochs 10–30)
Physics loss weight $\lambda_{\text{phy}}^{\text{max}}$	0.05
Mode consistency weight $\lambda_{\text{mode}}^{\text{max}}$	0.01
Teacher forcing decay epochs E_{tf}	75

- **CW propagation residual:** For each pair of consecutive predicted states, the norm of deviation from free CW evolution is computed as $\|s_{t+1} - \Phi(\Delta t) \cdot s_t\|$.
- **Δv magnitude distribution:** The model's Δv estimates are compared against the physically plausible bound of 3 m/s.

The CW residual statistics are: mean = $4.52 \times 10^{-4} \text{ km/s}$, standard deviation = $6.31 \times 10^{-4} \text{ km/s}$, and maximum = $1.73 \times 10^{-2} \text{ km/s}$. These small residual values indicate that the predicted trajectories closely satisfy CW propagation between consecutive one-second steps, confirming that the physics loss terms have successfully embedded the governing equations into the model's behavior.

The estimated Δv magnitudes have a mean of $1.27 \times 10^{-3} \text{ km/s}$ (1.27 m/s) and a maximum of $2.94 \times 10^{-3} \text{ km/s}$ (2.94 m/s), both within the prescribed 3 m/s bound. The distribution is concentrated below 2 m/s with a long tail approaching the limit, consistent with the PPO policy's action space that caps Δv at 3 m/s during data generation.

5.2.3. Ablation: impact of physics-informed components

To quantify the contribution of the physics-informed components, the full PI-LSTM model is compared against two ablated variants trained under identical conditions:

Table 3

BiGRU intention recognition model: architecture and training hyperparameters.

Parameter (Symbol)	Value
<i>Architecture</i>	
Physical feature channels d_{phys}	10
Global feature count d_{global}	9
Total input channels $d_{\text{in}}^{\text{rec}} = 6 + d_{\text{phys}}$	16
Conv1D kernel size k_{conv}	3
Conv1D output channels d_{conv}	128
BiGRU layers n_{gru}	2
BiGRU hidden size H_{gru}	128
Per-target embedding dim $d_{\text{emb}} = 2H_{\text{gru}}$	256
Attention heads n_{head}	4
Pairwise MLP hidden units $d_{\text{pair}}^{\text{hid}}$	128
Pairwise output dimension $d_{\text{pair}}^{\text{out}}$	256
Shared projection dimension d_{shared}	64
Number of N classes $K_N = N_{\text{max}} - N_{\text{min}} + 1$	3
Huber loss threshold δ_{huber}	1.0
Cosine penalty weight α_{cos}	0.1
Trainable parameters $N_{\text{param}}^{\text{rec}}$	$\approx 3.5 \times 10^6$
<i>Training</i>	
Optimizer	AdamW
Learning rate η_{rec}	10^{-3}
Weight decay $\lambda_{\text{wd}}^{\text{rec}}$	10^{-4}
Batch size B_{rec}	128
Dropout rate $p_{\text{drop}}^{\text{rec}}$	0.2
Gradient clipping ℓ_2 threshold g_{clip}	1.0
Max epochs E_{rec}	200
Early stopping patience P_{rec}	30
Scheduler initial period T_0	20
Scheduler multiplier T_{mult}	2
<i>Multi-task loss weights</i>	
N classification λ_N	1.0
d_{min} regression λ_d	1.0
ϕ regression λ_{ϕ}	2.0

Table 4

Trajectory prediction performance on the test set (3,568 samples). All metrics reported in original physical units.

Metric	Overall	Position	Velocity
MSE	1.72×10^{-2}	2.18×10^{-2}	7.09×10^{-6}
RMSE	1.31×10^{-1}	1.48×10^{-1}	2.66×10^{-3}
MAE	8.57×10^{-2}	9.65×10^{-2}	1.82×10^{-3}

1. **Standard LSTM (no physics):** The baseline encoder-decoder LSTM without the Δv estimator, condition builder, and physics loss terms.
2. **PI-LSTM without physics loss:** The PI-LSTM architecture with the Δv estimator and condition builder, but trained with only $\mathcal{L}_{\text{MSE}}$ throughout.

As shown in Table 5, the full PI-LSTM achieves the best position accuracy and the strongest physical consistency. The standard LSTM, despite achieving competitive MSE,

Table 5Ablation study on physics-informed components. CW residual and Δv metrics quantify physical consistency.

Model variant (m/s)	Pos. RMSE (km)	CW resid. (km/s)	Δv_{max}
Standard LSTM	0.162	9.81×10^{-4}	5.23
PI-LSTM (without phys. loss)	0.151	7.44×10^{-4}	3.87
PI-LSTM (full)	0.148	4.52×10^{-4}	2.94

produces Δv estimates exceeding the 3 m/s bound (maximum 5.23 m/s), indicating a lack of physical awareness. Adding the Δv estimator and condition builder without physics loss (second row) already improves both accuracy and consistency, confirming that the conditioning mechanism alone provides useful inductive bias. The addition of physics loss terms further tightens the CW residual and enforces the Δv bound, yielding predictions that are both more accurate and more physically plausible.

5.3. Intention recognition results

5.3.1. Overall performance

Table 6 reports the test-set performance of the BiGRU-based intention recognition model for all three prediction tasks.

Table 6

Intention recognition performance on the test set (3,568 samples).

Task & Metric	Value
<i>N — cluster size classification</i>	
Accuracy	99.1%
Macro F1	0.989
<i>d_{min} — minimum distance regression</i>	
MAE	0.073 km
RMSE	0.187 km
R^2	0.991
<i>ϕ — dispersion angle regression</i>	
MAE	0.284 rad
Circular MAE	0.261 rad
RMSE	0.357 rad
R^2	0.621

The model achieves near-perfect classification of cluster size (N accuracy $> 99\%$), indicating that the number of targets is reliably identifiable from the trajectory data alone. The minimum distance regression attains $R^2 = 0.991$, with a mean absolute error of only 73 m, demonstrating accurate capture of the geometric proximity information.

The dispersion angle ϕ is the most challenging task ($R^2 = 0.621$, MAE = 0.284 rad). This is expected because ϕ depends on identifying the nearest and farthest targets at a specific time step ($t_{d_{\text{min}}}$), requiring fine-grained spatial reasoning across multiple objects. The circular MAE of

0.261 rad (approximately 15 degrees) indicates reasonable angular resolution for operational intent assessment.

5.3.2. Cluster size classification analysis

The confusion matrix for N classification reveals that most misclassifications occur between adjacent classes ($N = 2 \leftrightarrow 3$ and $N = 3 \leftrightarrow 4$), with no cross-class errors between $N = 2$ and $N = 4$. The per-class F1 scores are 0.988 ($N = 2$), 0.986 ($N = 3$), and 0.989 ($N = 4$), confirming balanced performance despite the imbalanced class distribution.

5.3.3. Contribution of physics-informed features

To evaluate the impact of the physics-informed feature engineering, the full model (16 input channels: 6 states + 10 physical features) is compared against a reduced variant using only the raw 6-dimensional state vectors as input, while keeping the network architecture unchanged. Table 7 summarizes the results.

Table 7

Ablation study on physics-informed features for intention recognition.

Input features	N Acc.	$d_{\min}$ MAE (km)	ϕ MAE (rad)
6 raw states only	96.2%	0.214	0.411
6 states + 10 phys. features (full)	99.1%	0.073	0.284

The physics-informed features yield substantial improvements across all three tasks. The most dramatic gain is in $d_{\min}$ regression (MAE reduced by 66%, from 0.214 to 0.073 km), which directly benefits from the per-target distance and CW residual features. The ϕ regression improves by 31% (from 0.411 to 0.284 rad), attributable to the inter-target cosine angle statistics that provide direct geometric cues for angular prediction. These results confirm that explicitly injecting orbital dynamics knowledge through feature engineering is critical for achieving high performance on this moderate-size dataset.

5.4. Discussion

5.4.1. Complementarity of the two stages

Trajectory prediction and intention recognition serve complementary roles: the former provides dense future state estimates for short-term kinematic reasoning, while the latter extracts a compact semantic summary of the scene. In an operational pipeline, the predicted trajectory $\hat{\mathbf{X}}_{\text{next}}$ from Stage 1 could be fed as additional input to the intention recognition module, potentially improving its performance by providing a richer temporal context than the 10-step observation window alone. This extension is left for future work.

5.4.2. Role of physics in generalization

The physics-informed components — CW-based feature engineering, Δv conditioning, and physics loss regularization — are not merely performance enhancers on the training distribution; their primary value lies in constraining the model's behavior to remain physically plausible under distributional shift. A purely data-driven model trained on a specific maneuver regime may produce catastrophic predictions when encountering novel maneuver patterns. By embedding the governing equations, the approach provides a degree of extrapolation robustness essential for SSA applications where the range of possible behaviors is large relative to the available training data.

5.4.3. Multi-scenario dataset diversity

The inclusion of five distinct maneuver intention scenarios (pursuit, reconnaissance, ambush, obstruction, and mixed strategies) in the training data provides a level of behavioral diversity that is absent from single-scenario datasets. This diversity, achieved through different PPO reward structures, ensures that the learned representations are not overfitted to a narrow class of trajectories. The models trained on this multi-scenario dataset demonstrate consistent performance across all scenario types during evaluation, suggesting that the physics-informed features capture invariant kinematic patterns that transcend specific maneuver objectives.

5.4.4. Limitations

Several limitations warrant acknowledgment. First, the CW model is a linearization valid only when inter-satellite separations are much smaller than the orbital radius; its accuracy degrades for widely spaced clusters or highly eccentric orbits. Second, the dataset is generated from a single reference orbit (480 km altitude, near-circular), and the models' transferability to different orbital regimes has not been tested. Third, all maneuvers are modeled as instantaneous impulses, whereas real thrusters produce finite-duration burns. Fourth, the intention recognition task defines only three label types (N , $d_{\min}$, ϕ); operational scenarios may require richer intent taxonomies including specific maneuver type classification among the five scenarios and goal prediction.

6. Conclusions

This paper has presented a two-stage deep learning framework for maneuverable satellite cluster trajectory prediction and intention recognition. The first stage employs a physics-informed conditional LSTM that integrates Clohessy-Wiltshire orbital dynamics through a differentiable Δv estimation module, a condition vector builder that adapts the LSTM gating to the estimated maneuver intensity, and a physics-regularized loss function. The second stage uses a BiGRU-based permutation-invariant network with physics-informed feature engineering to perform multi-task intention recognition, classifying the number of cluster objects and regressing the minimum approach distance and spatial dispersion angle. The training dataset of 23,787

samples is generated via PPO-based simulations covering five maneuver intention scenarios: pursuit, reconnaissance, ambush, obstruction, and mixed strategies, providing behavioral diversity that enhances generalization.

Numerical simulations validate the effectiveness of both stages. The trajectory prediction model achieves a position RMSE of 0.148 km with strong CW self-consistency (mean residual 4.52×10^{-4} km/s) and Δv estimates bounded within the physically plausible range. The intention recognition model attains 99.1% accuracy on N classification, an R^2 of 0.991 on $d_{\min}$ regression, and an R^2 of 0.621 on ϕ regression.

The key findings of this study are summarized as follows:

- 1. Physics injection is critical for sample efficiency.** On a moderate-size dataset (23,787 samples), both physics-informed loss terms (trajectory prediction) and physics-guided feature engineering (intention recognition) yield significant improvements over purely data-driven baselines. Ablation studies confirm that embedding domain knowledge through multiple complementary mechanisms — input conditioning, loss regularization, and feature construction — is more effective than any single mechanism alone.
- 2. Permutation-invariant architectures with explicit pairwise modeling are effective for multi-agent analysis.** The combination of a shared-weight per-target encoder, set attention aggregation, and a pairwise interaction module correctly handles variable numbers of targets while providing the relational features necessary for geometry-sensitive predictions such as the dispersion angle.
- 3. Multi-scenario training data enhances robustness.** The five distinct maneuver intention scenarios generated by different PPO reward structures produce a diverse training set that improves the models' ability to generalize across different operational regimes, compared to training on any single scenario alone.

Future research directions include: (1) extending the framework to eccentric orbits and non-Keplerian dynamics where the CW linearization no longer holds, e.g., by incorporating a differentiable numerical propagator based on the Tschauner–Hempel equations; (2) generalizing to a broader taxonomy of maneuver types beyond the Δv magnitude characterization, including explicit classification among the five intention scenarios studied in this work; (3) investigating few-shot or meta-learning approaches for rapid adaptation to novel orbital regimes with limited data; (4) exploring the use of the predicted trajectory as an additional input modality for intention recognition, potentially closing the loop between the two stages; and (5) validating the approach on real-world space surveillance data, where measurement noise, sparse observations, and uncertain maneuver timing introduce additional challenges not present in simulation.

CRedit authorship contribution statement

Xuanyu Luo: Conceptualization, Methodology, Software, Writing – original draft. **Chuang Liu:** Formal analysis, Validation, Writing – review & editing. **Xiaokui Yue:** Data curation, Investigation, Visualization.

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